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COPULA ACTIVE SUBSPACES II: ERROR DECOMPOSITION, A POSTERIORI ESTIMATION, AND SHARPNESS OF THE BOUNDS*

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Abstract.
The Copula Active Subspaces (CAS) method of Part I estimates a non-Gaussian observation-noise density $\pi_{\boldsymbol{n}}$ from residual samples. Its KL divergence $\mathcal{D}_{\mathrm{KL}}(\pi_{\boldsymbol{n}} \,\|\, \widehat{\pi}_{\boldsymbol{n}})$ admits an exact decomposition into marginal, projection and reduced-density terms and bounds the posterior KL divergence averaged over the true predictive distribution. We derive a priori bounds for these terms, together with estimates or bounds from residual samples. Weighted nearest-neighbour entropy estimates are used for the marginal and reduced-density terms, with asymptotic standard errors in exact rank coordinates. For the projection error, the score is fitted on one subsample, and its mean squared magnitude outside the retained subspace is estimated on the other. Corrections for score and rank-transform errors then give a theoretical upper bound with probability at least $1-\delta$. Tests on three known noise laws in dimension $d = 20$ use the estimated rank transform. In 11 of 12 combinations of example and sample size, the sum of the two estimated terms and the trace bound exceeds the reference divergence after averaging over the estimated models. In these tests, score-estimation error is the largest contribution to looseness. The overestimate from the Gaussian log-Sobolev inequality alone is at most a factor of 1.98.

Key words. copula, active subspace, score matching, dimension reduction, Hermite polynomials, Bayesian inference problems, non-Gaussian observation error

MSC codes. 62G07, 62G20, 62H05, 62B10, 60E15, 62F15

1. Introduction. Approximating the observation-noise density changes the posterior in a Bayesian inference problem. The noise-density KL divergence bounds the resulting posterior KL divergence, averaged over the true predictive distribution. Part I [10] estimates a non-Gaussian noise density $\pi_{\boldsymbol{n}}$ from residual samples, and tests the estimate against known noise laws. Such a comparison is unavailable in applications. We therefore consider the error

$$\mathcal{D}_{\mathrm{KL}}(\pi_{\boldsymbol{n}} \,\|\, \widehat{\pi}_{\boldsymbol{n}}), \tag{1.1}$$

of the CAS estimate $\widehat{\pi}_{\boldsymbol{n}}$, estimating its individual contributions from the residual samples.

The decomposition in Proposition 2.1 assigns one error term to each estimation step of CAS: (1.1) equals $\mathcal{E}_{\mathrm{marg}} + \mathcal{E}_{\mathrm{proj}} + \mathcal{E}_{\mathrm{dens}}$. The marginal term $\mathcal{E}_{\mathrm{marg}}$ consists of d one-dimensional divergences and two corrections; $\mathcal{E}_{\mathrm{dens}}$ is a divergence between r-dimensional laws. For the projection error $\mathcal{E}_{\mathrm{proj}}$, the rank-$r$ reduction admits a bound in terms of the copula-score variance outside the retained subspace. This is the Gaussian-reference bound of certified dimension reduction [36].

A direct entropy estimate of (1.1) would involve the differential entropy of $\pi_{\boldsymbol{n}}$ in dimension d. For densities supported on $[0,1]^d$ in Lipschitz balls of smoothness $s = 2$, the minimax root-mean-square error has order $(N \log N)^{-2/(d+2)} + N^{-1/2}$ [19,

*This is Part II of a two-part work. The companion paper *Copula Active Subspaces I: A Score-Covariance Method for Reduced-Order Non-Gaussian Density Estimation* defines the estimator analyzed here.

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Thm. 1]. The decomposition reduces the required entropy calculations to dimensions 1 and r. No entropy estimate is needed for the bound on $\mathcal{E}_{\mathrm{proj}}$.

The a priori bounds in Section 3 depend on the rank r, the multi-index sets and the sample size N. Normalizing the fitted joint density introduces a correction in the marginal bound. The bound minus this correction tends to zero as $N \to \infty$, under the stated assumptions on the marginal estimates and rank-transform error. The bounds on $\mathcal{E}_{\mathrm{proj}}$ and $\mathcal{E}_{\mathrm{dens}}$ keep approximation, regularization and constraint terms that do not vanish as $N \to \infty$. In the examples of Section 6, $\mathcal{E}_{\mathrm{proj}}$ is the largest term for $N \geq 12{,}500$, and the Gaussian log-Sobolev bound overestimates it by a factor between 1.64 and 1.98. We also apply gradient descent to the tighter bound of [24]. The subspace moves by $\|\sin\Theta\|_F \leq 0.0004$ from the top-r eigenbasis of $\boldsymbol{C}$ used by CAS. Most of the remaining looseness of the a priori bound on these examples is due to the Hermite truncation of the score (Subsection 6.4).

For the a posteriori analysis in Section 4, we estimate $\mathcal{E}_{\mathrm{marg}}$ and $\mathcal{E}_{\mathrm{dens}}$ by weighted nearest-neighbour entropy estimators. Under the stated assumptions these attain the parametric rate in exact rank-Gaussianized coordinates. To bound $\mathcal{E}_{\mathrm{proj}}$, we fit the score and rank transform on half of the sample. The other half estimates the mean squared magnitude of the fitted score outside the retained subspace. Under the stated assumptions, a theoretical upper bound for this quantity holds with probability at least $1-\delta$. It becomes a bound on $\mathcal{E}_{\mathrm{proj}}$ after adding corrections for errors in the score estimate and rank transform. The increment between two nested multi-index sets estimates the Hermite truncation error under a saturation assumption. It does not estimate the sampling or regularization errors.

Section 2 defines the estimator and decomposes its error. The three numerical examples are reported in Section 6.

2. The estimator and its exact error decomposition. The two stages of CAS estimate the density $\pi_{\boldsymbol{n}}$ of the noise $\boldsymbol{n} \in \mathbb{R}^d$ from N residual samples (Part I [10], Algorithm 3.1). For marginal densities π_i with CDFs F_i, the componentwise map $\mathcal{Z}(\boldsymbol{x})_i := \Phi^{-1}(F_i(x_i))$ produces standard Gaussian marginals (2.1), with Φ the standard normal CDF. The transformed noise $\boldsymbol{Z} := \mathcal{Z}(\boldsymbol{n})$ has law $\pi_{\boldsymbol{Z}}$. Replacing F_i by the rank-based $\widehat{F}_i$ of Part I defines $\widehat{\mathcal{Z}}$. The marginal densities are estimated separately by Gaussian-kernel estimates $\widehat{\pi}_i$ [31].

Let γ_k denote the standard Gaussian density on $\mathbb{R}^k$. The copula density is $c^Z := \pi_{\boldsymbol{Z}}/\gamma_d$. Its score $\mathcal{S} := \nabla \log c^Z$ has second-moment matrix $\boldsymbol{C} := \mathbb{E}_{\pi_{\boldsymbol{Z}}}[\mathcal{S}\mathcal{S}^\top]$ (2.2), referred to as the score covariance. With eigenvalues ordered as $\lambda_1(\boldsymbol{C}) \geq \cdots \geq \lambda_d(\boldsymbol{C})$, a top-$r$ eigenbasis $\boldsymbol{V}_r^{\boldsymbol{C}}$ spans the copula active subspace.

Stage 1 fits the copula score by Hermite score matching [22] in the span of the gradients of multivariate Hermite polynomials H_α. The polynomials are orthonormal in $L^2(\gamma_d)$ and indexed by $\Lambda := \Lambda_{K_1,q_1}$. The set $\Lambda_{K,q}$ contains indices with $1 \leq |\alpha|_1 \leq K$ and at most q nonzero entries, except for the linear indices e_i (3.2). This fit defines $\widehat{\mathcal{S}} := \nabla \sum_{\alpha\in\Lambda} \widehat{\theta}^d_\alpha H_\alpha$ (3.3), with coefficient vector $\widehat{\boldsymbol{\theta}}^d := (\widehat{\theta}^d_\alpha)_{\alpha\in\Lambda}$. From the empirical second-moment matrix $\widehat{\boldsymbol{C}}$ of $\widehat{\mathcal{S}}$ (3.5), we choose a top-r eigenbasis $\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}}$. The superscript distinguishes this basis from a principal-component basis of $\boldsymbol{Z}$.

A basis $\boldsymbol{V}_r \in \mathrm{St}(r,\mathbb{R}^d)$ and its orthonormal completion $\boldsymbol{V}_\perp$ separate the retained coordinates $\boldsymbol{U} := \boldsymbol{V}_r^\top \boldsymbol{Z}$ from the discarded coordinates $\boldsymbol{W} := \boldsymbol{V}_\perp^\top \boldsymbol{Z}$. The conditional law of $\boldsymbol{W}$ given $\boldsymbol{U}$ is denoted by $\pi_{\boldsymbol{W}|\boldsymbol{U}}$. Define the reduced factor $c_r^U := \pi_{\boldsymbol{U}}/\gamma_r$ and its logarithm $\mathcal{L}_r := \log c_r^U$. Part I (Subsection 2.1) uses this factor to define the reduced model $\pi_{\boldsymbol{n}}(\boldsymbol{x};\boldsymbol{V}_r) := c_r^U(\boldsymbol{V}_r^\top \mathcal{Z}(\boldsymbol{x}))\prod_i \pi_i(x_i)$ (2.4). For the fitted estimator, we use $\boldsymbol{V}_r = \widehat{\boldsymbol{V}}_r^{\boldsymbol{C}}$ in these definitions.